Woosik Kim

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EXPERIENCE

LG Uplus **Jun 2026 - Jul 2026**
AI Red Team Intern, Information Security Center

- Architected a **7-stage** AI red team lifecycle for a major South Korean telecom carrier to secure Shadow AI company-wide.
- Defined AI asset discovery across **5** detectors (DNS, EDR, OAuth consent, secret scans, cloud audit logs).
- Built a **19-scenario** telecom attack taxonomy mapped to MITRE ATLAS and OWASP LLM Top **10**, verified with canary tokens.
- Designed AVS, a CVSS-based AI risk score adding agency and exploit rate, and defended it to the CTO and security chief.

Match-a **Jun 2026 - Present**
Tech Lead (Software Engineering Intern)

- Secured the app with Twilio webhook signature verification, LLM route rate limits, and fail-fast secret validation.
- Added Semgrep SAST and gitleaks secret scanning to CI and cleared high-severity npm advisories.
- Engineered the GitHub Actions CI pipeline gating every merge on type checks, linting, a **340+** test suite, and a build.
- Instrumented Sentry error monitoring across server, edge, and client runtimes covering **80+** API routes.
- Lead **3** engineering interns at an early-stage consumer social platform, setting technical direction and reviewing their PRs.

Youngduk Electronics **May 2017 - May 2020**
Operations Technology Assistant

- Automated inventory and ERP reporting for an electronics distributor, cutting stock errors **30%** and raising data accuracy **20%**.

RESEARCH & PROJECTS

Coverage Keeper | AI Medicaid Letter Reader | 2nd Place, Healthcare Track, Solix Empower Hackathon **Oct 2026**
Team of 3 | *Next.js, TypeScript, Anthropic API, zod* | github.com/AdityaJadhav17/coverage-keeper

- Protected health data with upload type, size, and rate limits, guardrails on every output field, and content-free logs.
- Engineered the letter-reading backend: vision LLM calls with schema-validated output, **429/5xx** retry, and a safe fallback.
- Curated a **13-notice** labeled eval set with a fixed dev/test split, reading **4 of 4** held-out deadlines correctly.

WatchTower | Self-hosted Browser Monitoring SDK (Team Project) **Apr 2026 - Jun 2026**
Live demo: cse110-sp26-group09.github.io/Watchtower-Course-Project

- Found and fixed a stored XSS vulnerability in the monitoring dashboard by escaping untrusted event data in HTML output.
- Added request validation and **37+** server unit tests to the Node.js event API, plus Playwright end-to-end tests in CI.

ShowBrand: AI Reliability Research | Faculty Mentor Program, UC San Diego **Jan 2026 - May 2026**
Advised by Prof. Pengtao Xie (ECE) | *Presented at MURALS 2026* | github.com/woosik-study/ShowBrand-Product

- Stress-tested a vision LLM against blur, logo occlusion, and AI fakes in **4** controlled experiments on **136** images.
- Found an overconfidence failure: with logos occluded, brand accuracy fell **17 points** while model confidence fell only **0.7**.
- Shipped the findings into the app as confidence thresholds, a low-confidence retake prompt, and risk-based wording.

TECHNICAL SKILLS

Security: AI red teaming, threat modeling, MITRE ATLAS, OWASP Top **10** / LLM Top **10**, SAST (Semgrep), secret scanning (gitleaks)
Languages: Python, TypeScript/JavaScript, C/C++, SQL
Web & Backend: React, Next.js, Node.js, REST API design, PostgreSQL (Supabase)
AI & ML: LLM APIs (Anthropic, Vertex AI/Gemini), vision language models, structured output, LLM evaluation, uncertainty analysis
DevOps & Tooling: Linux, bash shell, Git, GitHub Actions CI/CD, Vercel, Sentry, Playwright

EDUCATION

University of California, San Diego **Expected Dec 2027**
Bachelor of Science in Computer Science
Relevant Coursework: Computer Systems & Architecture, Software Engineering, AI & Search, Statistical Learning

LEADERSHIP, ACTIVITIES & AWARDS

Military Service: Team Leader, Republic of Korea Army (**2020** - **2022**), led and trained a unit of **36** recruits
Activities: Web Developer, TESC | Developer, Likelion UCSD | IDEA Scholar, Jacobs School of Engineering | Member, Council on Undergraduate Research (CUR) | Math & Programming Mentor, MESA
Awards: STEM Math Achievement Award (**2025**), Bill and Mary Kim Scholarship (**2024**, **2025**) | **Languages:** English, Korean